

CSE 331

AUTOMATA AND COMPUTABILITY

Lecture 6

Spring 2026

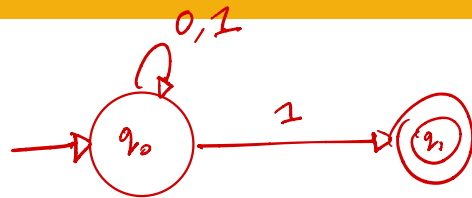
Today's learning goals

Sipser Ch 1.2, 1.3

- Design NFA recognizing a given language
- Defining NFA in closure constructions formally.
- Convert an NFA (with or without spontaneous moves) to a DFA recognizing the same language

Acceptance in an NFA

$$\Sigma = \{0, 1\}$$



An NFA $(Q, \Sigma, \delta, q_0, F)$ accepts a string w in Σ^* iff we can write $w = y_1 y_2 \cdots y_m$ where each $y_i \in \Sigma_\varepsilon$ and **there is** a sequence of states $r_0, \dots, r_m \in Q$ such that

1. $r_0 = q_0$

$$\Sigma = \underline{\Sigma} \cup \varepsilon$$

2. $r_{i+1} \in \delta(r_i, y_{i+1})$ for each $i = 0, \dots, m-1$

3. $\underline{r_m} \in F$.
 ↓ State ↓ Symbol

$$w = 0 \underline{1} 1$$

$$y_1 \quad y_2 \quad y_3$$

$$\delta(r_i, 1) = \{r_1, r_2\}$$

$$\parallel$$

$$r_i$$

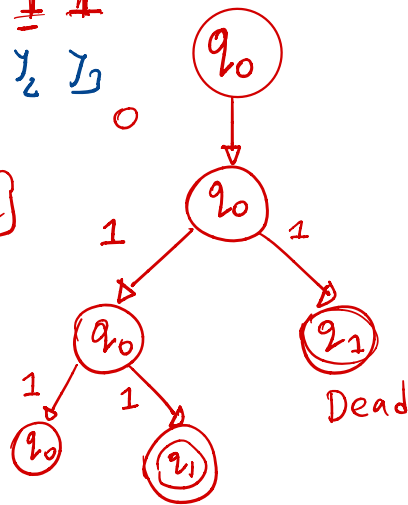
$$\underbrace{\hspace{1cm}}_{r_{i+1}}$$

$$|Q| = 2$$

$$|P(Q)| = 2^2$$

$$Q \times \Sigma_\varepsilon \rightarrow P(Q)$$

$$\{q_0, q_1\} \times \Sigma_\varepsilon \rightarrow \{\{q_1\}, \{q_0\}, \{q_0, q_1\}, \{\}\}$$



More differences between NFA and DFA

- **DFA**: unique computation path for each input
- **NFA**: allow several (or zero) alternative computations on *same input*
 - $\delta(q,x)$ may specify **more than one** possible next states
 - ϵ transitions allow the machine to **transition between states spontaneously**, without consuming any input symbols
- Types of components of formal definition
 - **DFA** $\delta : Q \times \Sigma \rightarrow Q$
 - **NFA** $\delta : Q \times \Sigma_{\epsilon} \rightarrow \mathcal{P}(Q)$

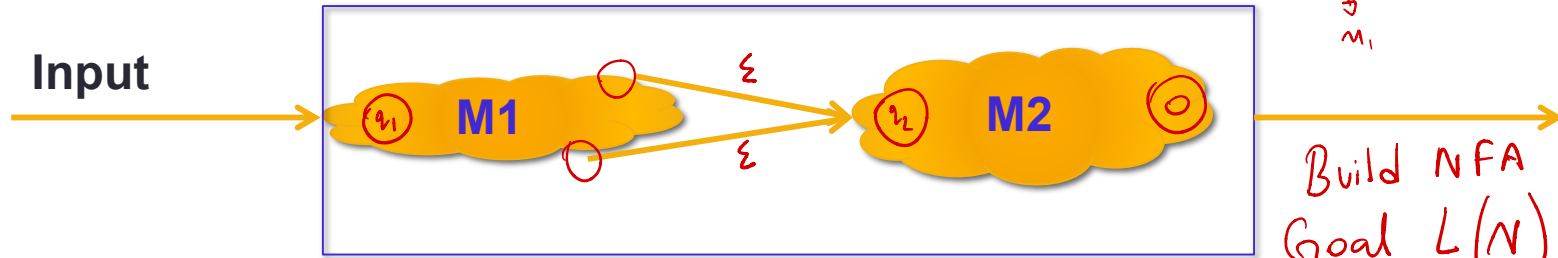
Similarities between DFA and NFA

- If L is a language recognized by a DFA, is there some NFA that recognizes it?

- A. Yes
- B. No
- C. Depends on L
- D. I don't know.

Concatenation

- "Guess" some stage of input at which switch modes



$$w = xy$$

$$x \in A_1 \quad y \in A_2$$

$\downarrow \quad \downarrow$
 $m_1 \quad m_2$

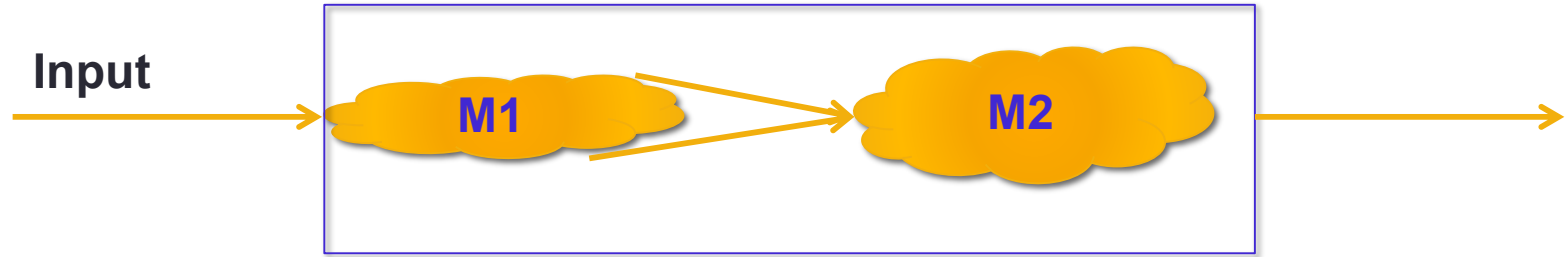
Build NFA
Goal $L(N) = L(M_1) \cdot L(M_2)$
 $= A_1 \cdot A_2$

Given DFAs $M1 = (Q1, \Sigma, \delta1, \underline{q1}, F1)$, $M2 = (Q2, \Sigma, \delta2, q2, F2)$

build NFA $N = (Q1 \cup Q2, \Sigma, \underline{\delta}, q1, F2)$ with $\delta \dots$

$$Q = Q_1 \cup Q_2 = \{ \dots \}$$

Concatenation



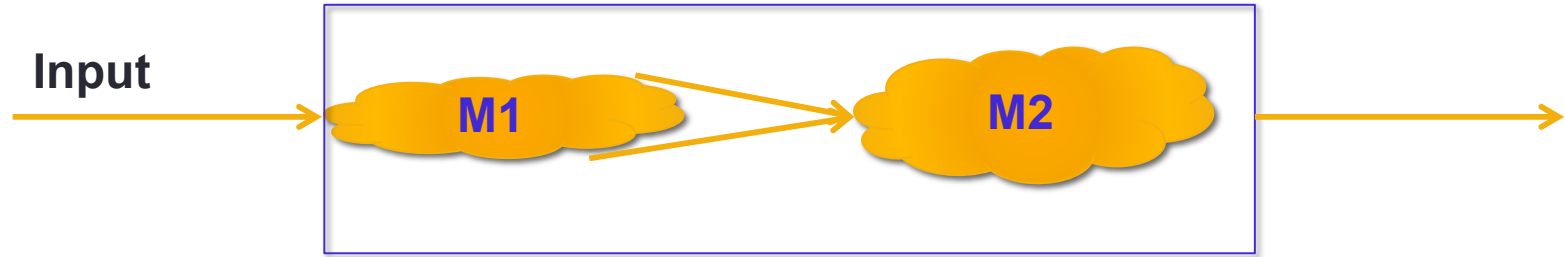
$$\delta(q, x) = \begin{cases} \delta_1(q, x) \\ \delta_2(q, x) \end{cases}$$

if q is in Q1, x is in Σ ✓

if q is in Q2, x is in Σ

?

Concatenation

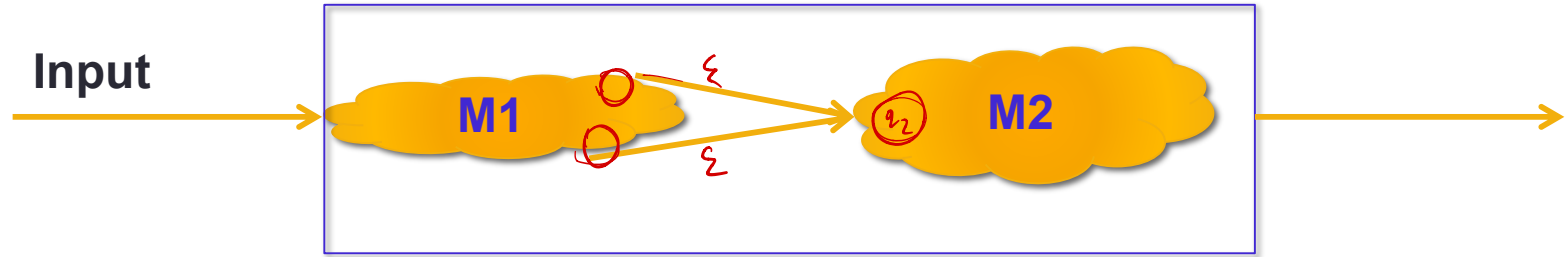


$$\delta(q, x) = \{\delta_1(q, x)\}$$
$$\{\delta_2(q, x)\}$$

if q is in Q_1 , x is in Σ

if q is in Q_2 , x is in Σ

Concatenation



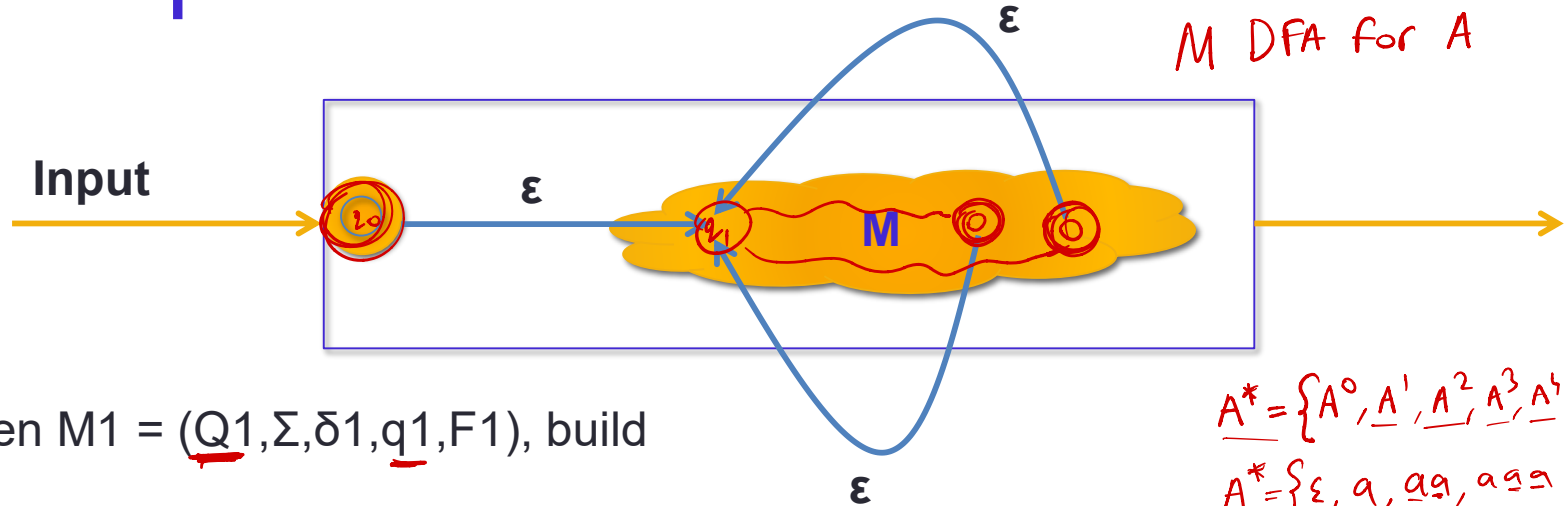
$$\delta(q, x) = \begin{cases} \delta_1(q, x) \\ \delta_2(q, x) \\ \{q_2\} \\ \emptyset \end{cases}$$

if q is in Q_1 , x is in Σ
 if q is in Q_2 , x is in Σ
 if q is in F_1 , $x = \epsilon$
otherwise

Correctness proof in the book (page 61)

Star operation

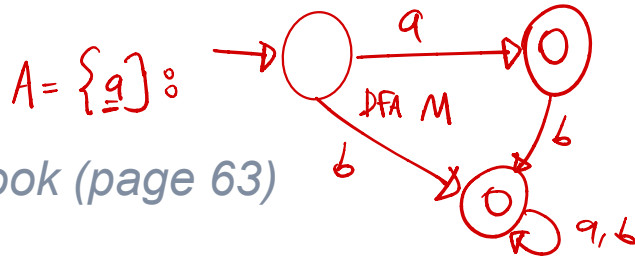
$$A^* = \{x_1 x_2 \dots x_k : k \geq 0, x_i \in A\}$$



Given $M_1 = (\underline{Q_1}, \underline{\Sigma}, \underline{\delta_1}, \underline{q_1}, \underline{F_1})$, build

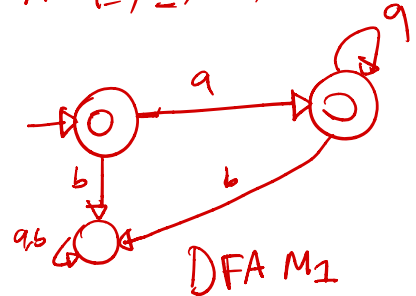
$$M = (Q_1 \cup \{q_0\}, \Sigma, \delta, q_0, F_1 \cup \{q_0\})$$

and $\delta(q, x) = \dots$



$$A^* = \{A^0, A^1, A^2, A^3, A^4, \dots\}$$

$$A^* = \{\epsilon, a, aa, aaaa, \dots\}$$

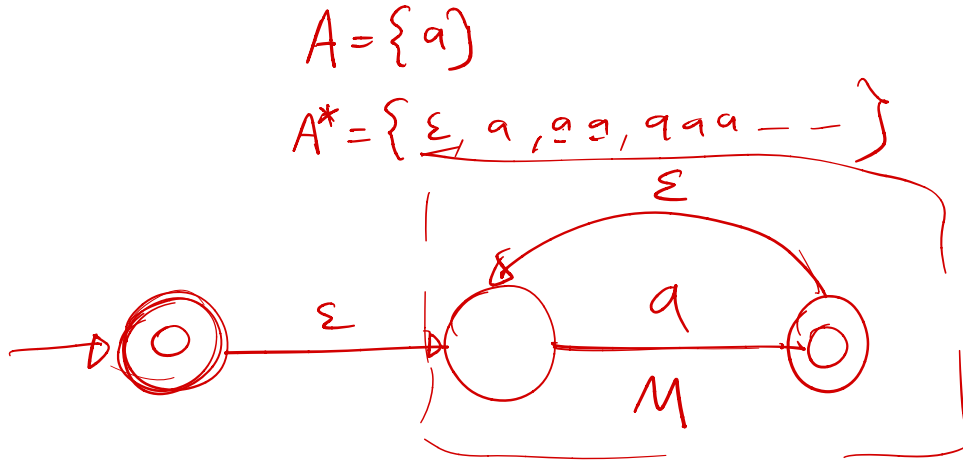


Construction in the book (page 63)

Next steps

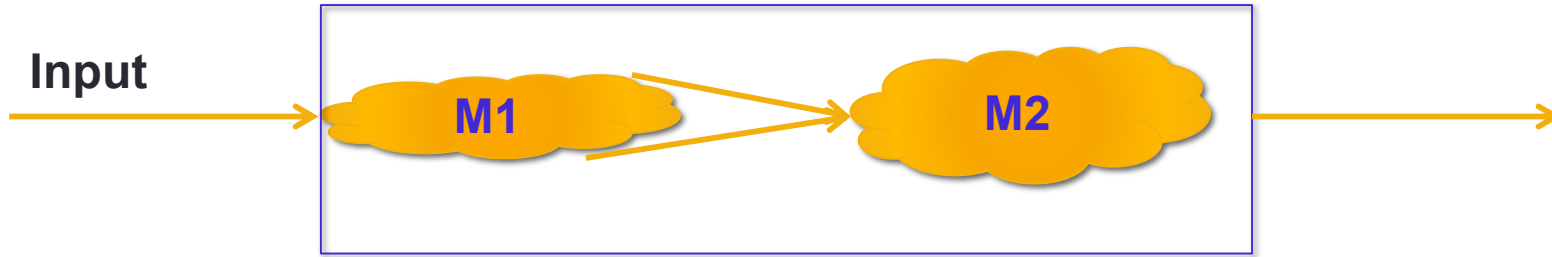
- Showing that NFA and DFA are equally expressive.

$a^* \cup b^*$

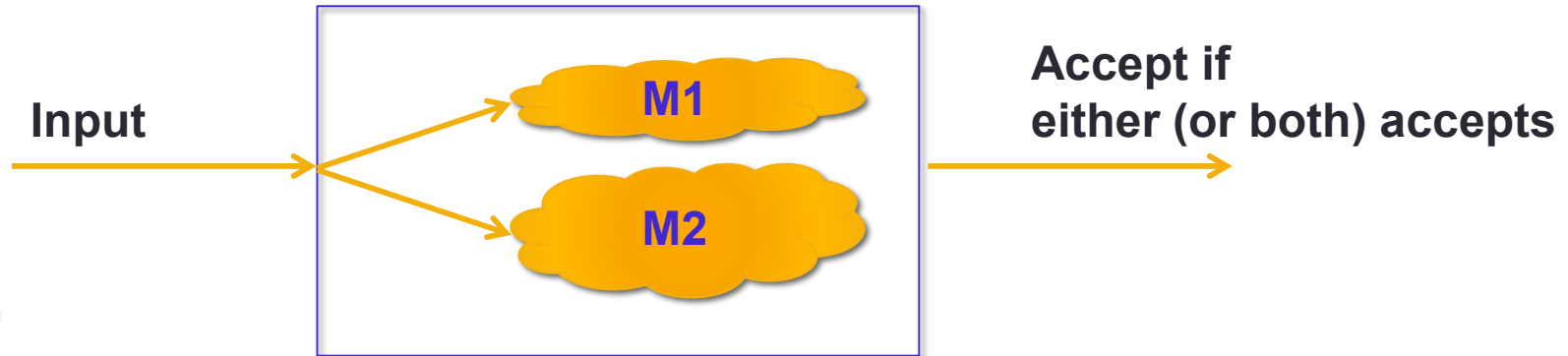


Nondeterministic finite automata

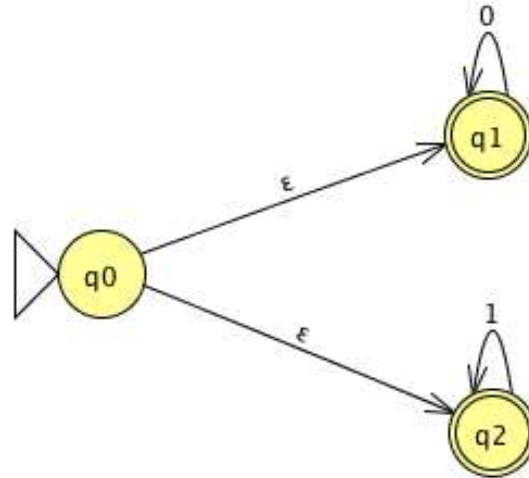
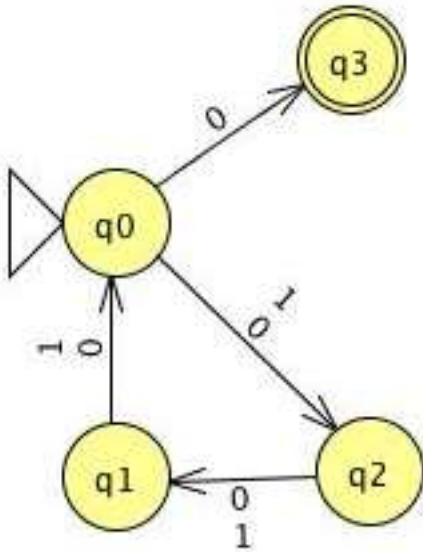
- "Guess" some stage of input at which switch modes



- "Guess" one of finite list of criteria to meet



NFA



Simulating NFA with DFA

Not quite a closure proof, but ...

Proof:

Given name variables for sets, machines assumed to exist.

WTS state goal and outline plan.

Construction using objects previously defined + new tools working towards goal. Give formal definition and explain.

Correctness prove that construction works.

Conclusion recap what you've proved.

Simulating NFA with DFA

For any language recognized by an NFA, there is a DFA that recognizes this language.

Proof:

Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

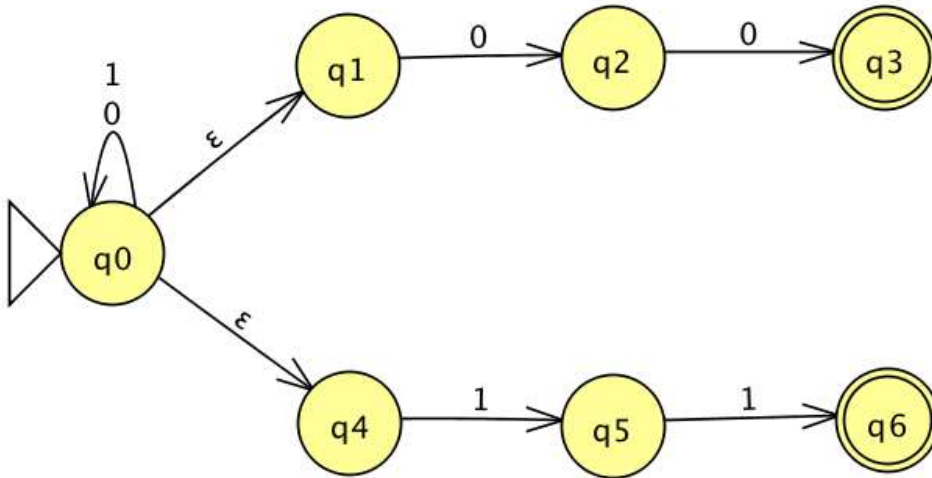
WTS there exists a DFA M with $L(M) = A$

Construction

Correctness

Conclusion

From NFA to DFA



Which of the following strings is accepted by this NFA?

- A. The empty string
- B. 01
- ☒ C. 111
- D. 0
- E. More than one of the above.

Subset construction

$$|Q| = n$$
$$|Q'| = 2^n$$

Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

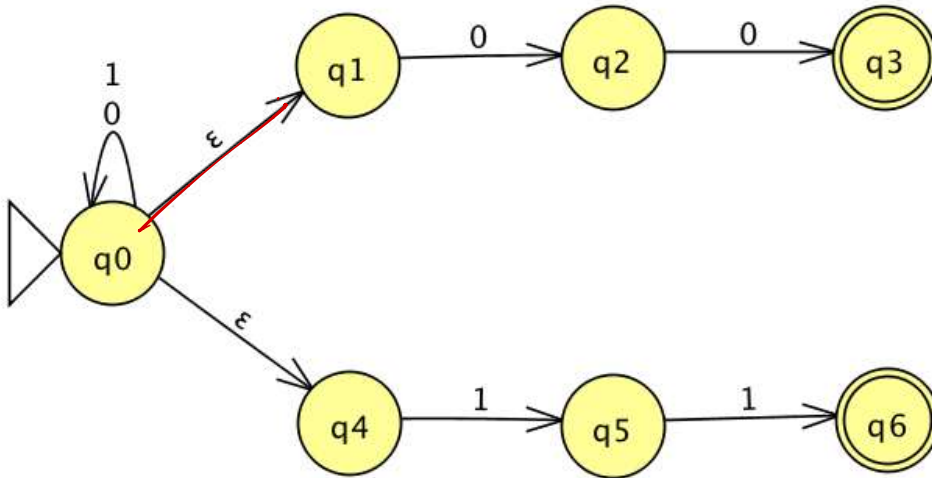
WTS there exists a DFA M with $L(M) = A$

Construction Define $M = (\underline{Q'}, \Sigma, \delta', \underline{q_0'}, F')$ with

- Q' = the power set of $Q = \{ \cancel{X} \mid X \text{ is a subset of } Q \}$
- $q_0' = \{ \text{states } N \text{ can be in before first input symbol read} \}$
- $F' = \{ \quad \quad \quad \}$
- $\delta' (\quad \quad \quad) =$

$$Q \times \Sigma \rightarrow \boxed{P(Q)}$$
$$\{ \text{---} \}$$

From NFA to DFA



$$\epsilon\text{-Closure}(q_0) = \{q_0, q_1, q_4\}$$

Start state in DFA

Which states can this NFA be in before first input symbol is read?

- A. q_0
- B. any state
- C. q_0, q_1
- D. q_0, q_4
- ☒ E. q_0, q_1, q_4

Subset construction

Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

WTS there exists a DFA M with $L(M) = A$

Construction Define $M = (Q', \Sigma, \delta', q_0', F')$ with

- $Q' =$ the power set of $Q = \{ X \mid X \text{ is a subset of } Q \}$
- $q_0' = \{ q_0 \} \cup \delta((q_0, \epsilon)) \dots = \{q_0\} \cup \{q_1, q_2\} = \{q_0, q_1, q_2\}$
- $F' = \{ \quad \quad \quad \}$
- $\delta' (\quad \quad \quad) =$

Subset construction

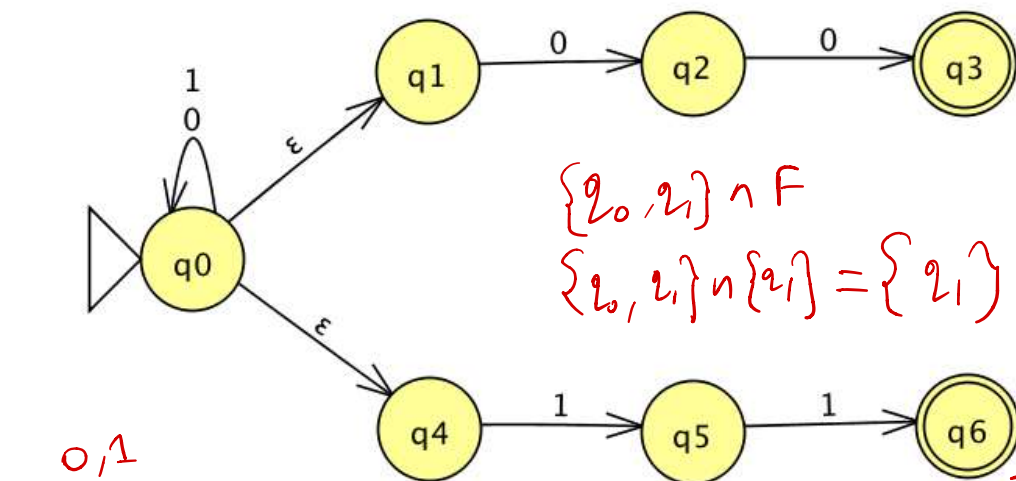
Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, \underline{F})$

WTS there exists a DFA M with $L(M) = A$

Construction Define $M = (Q', \Sigma, \delta', q_0', F')$ with

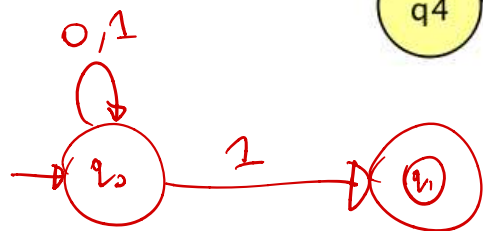
- $Q' =$ the power set of $Q = \{ \underline{X} \mid X \text{ is a subset of } Q \}$
- $q_0' = \{ q_0 \} \cup \delta((q_0, \epsilon)) \dots$
- $F' = \{ \text{guarantee at least one computation is successful} \}$
- $\delta' (\quad) =$

From NFA to DFA



$$\{q_0, q_1\} \cap F$$

$$\{q_0, q_1\} \cap \{q_1\} = \{q_1\}$$



$$Q' = P(Q) = \{\{q_0\}, \{q_1\}, \{q_0, q_1\}, \{q_3\}\}$$

$$F = \{q_1\}$$

Final state

What does it mean for a set of states X to guarantee at least one computation is successful?

- A. X is a subset of F
- B. $X = F$
- ☒ C. $X \cap F$ is nonempty
- D. X is an element of F
- E. None of the above.

$$\{q_1\} \cap \{q_1\} = \{q_1\}$$

Subset construction

Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

WTS there exists a DFA M with $L(M) = A$

Construction Define $M = (Q', \Sigma, \delta', q_0', F')$ with

- $Q' =$ the power set of $Q = \{ X \mid X \text{ is a subset of } Q \}$
- $q_0' = \{ q_0 \} \cup \delta((q_0, \epsilon)) \dots$
- $F' = \{ X \mid X \text{ is a subset of } Q \text{ and } X \cap F \text{ is nonempty} \}$
- $\delta' ((X, x)) = \{ \quad \quad \quad \}$

Subset construction

Given A, a language recognized by NFA $N = (Q, \Sigma, \delta, q_0, F)$

WTS there exists a DFA M with $L(M) = A$

Construction Define $M = (Q', \Sigma, \delta', q_0', F')$ with

- $Q' = \text{the power set of } Q = \{ X \mid X \text{ is a subset of } Q \}$
- $q_0' = \{ q_0 \} \cup \delta((q_0, \epsilon))$... $Q = \{ \underbrace{\{q_0\}}_{r_0}, \underbrace{\{q_1\}}_{r_1}, \underbrace{\{q_0, q_1\}}_{r_0, r_1}, \dots \}$
- $F' = \{ X \mid X \text{ is a subset of } Q \text{ and } X \cap F \text{ is nonempty} \}$
- $\delta'((X, x)) = \{ q \in Q \mid q \text{ is in } \delta(r, x) \text{ for some } r \text{ in } X \text{ or accessible via spontaneous moves} \}$

Made with Goodnotes

State in PFA
 $X \subseteq Q$

Types?

3

$\delta'((X, x)) = \{q \in Q \mid q \text{ is in } \delta(r, x) \text{ for some } r \in X \text{ or accessible via spontaneous moves}\}$

$X \in \Sigma$

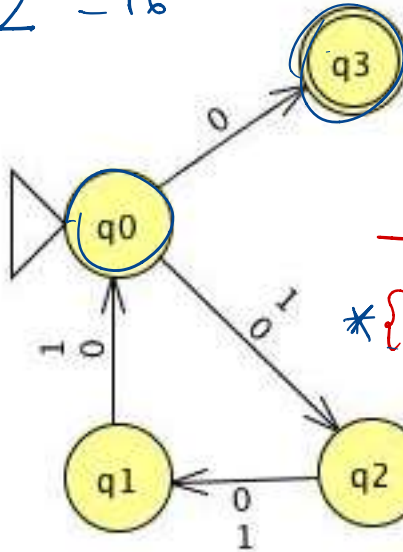
Subset construction examples

$$2^4 = 16$$

Without Epsilon

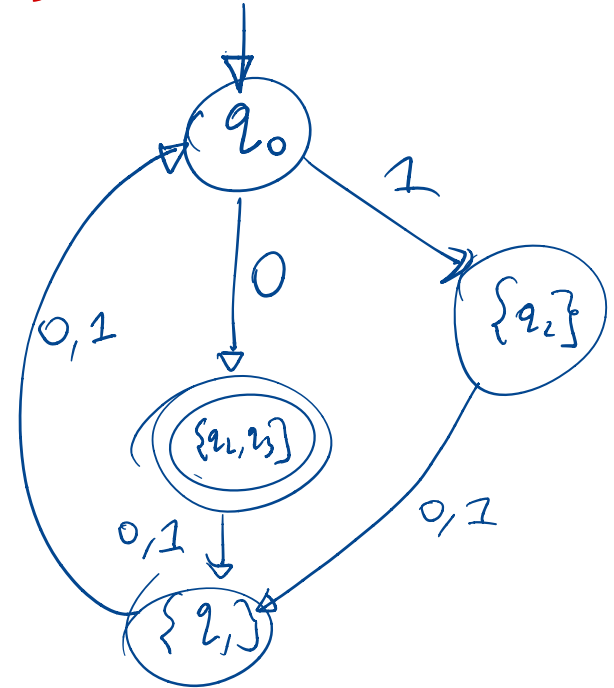
$$Q \times \Sigma \rightarrow Q$$

DFA M , $\Sigma = \{0, 1\}$



Σ	0	1
$\rightarrow q_0$	$\{q_2, q_3\}$	$\{q_2\}$
$*\{q_2, q_3\}$	$\{q_1\}$	$\{q_1\}$
$\{q_2\}$	$\{q_1\}$	$\{q_1\}$
$\{q_1\}$	$\{q_0\}$	$\{q_0\}$

$$X \cap F = \{q_2, q_3\} \cap \{q_3\} = \{q_3\}$$



Subset construction examples

$L(M_1) = \text{"all 0's or all 1's"}$

$L(M_1) = L(0^* \cup 1^*)$



start state.
 $\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$
 $\epsilon\text{-closure}\{q_1\} = \{q_1\}$
 $\epsilon\text{-closure}\{q_2\} = \{q_2\}$

Old state $\times \Sigma \rightarrow \epsilon\text{closure of new state.}$

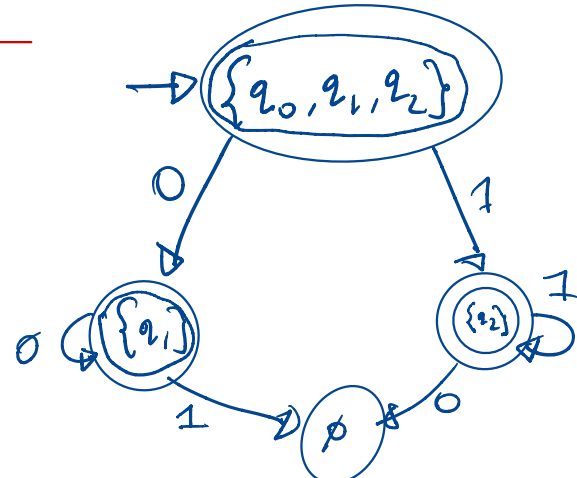
$q_0 = q_0$
 $\Sigma = \{0, 1\}$
 $F = \{q_1, q_2\}$
 $Q = \{q_0, q_1, q_2\}$

M_1

~~$\{q_0, q_1, q_2\}$~~
 $\{q_1\}$
 $\{q_2\}$

	0	1
$\{q_0, q_1, q_2\}$	$\{q_1\}$	$\{q_2\}$
$\{q_1\}$	$\{q_1\}$	$\emptyset$
$\emptyset$	$\emptyset$	$\{q_2\}$
$\{q_2\}$		

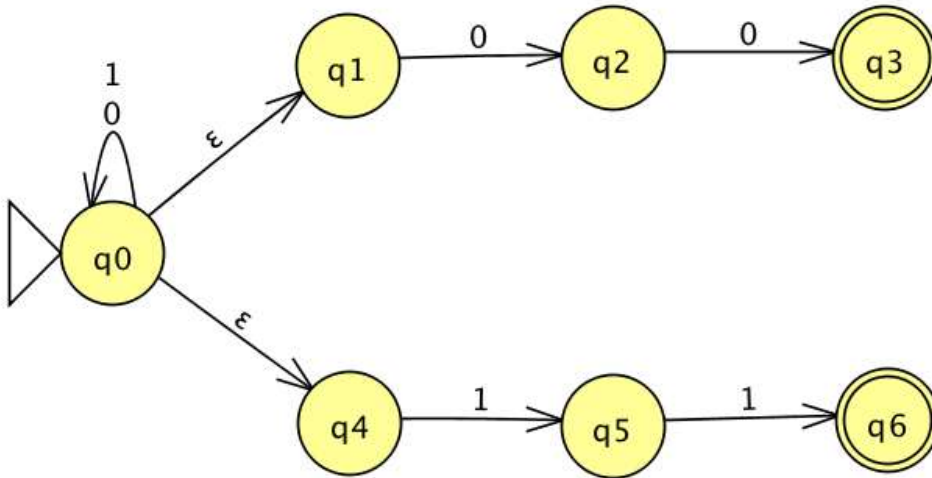
$\emptyset \rightarrow \text{lead state.}$



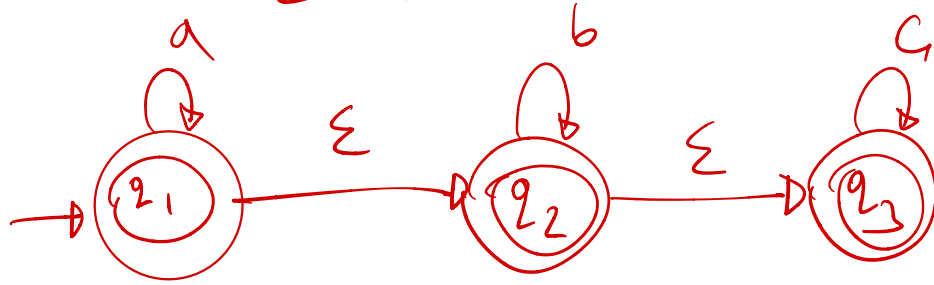
δ :

State / label	q_0	q_1	q_2
0	$\emptyset$	$\{q_1\}$	$\emptyset$
1	$\emptyset$	$\emptyset$	$\{q_2\}$
ϵ	$\{q_0, q_1, q_2\}$	$\emptyset$	$\emptyset$

Subset construction examples



$$\Sigma = \{a, b, c\}$$

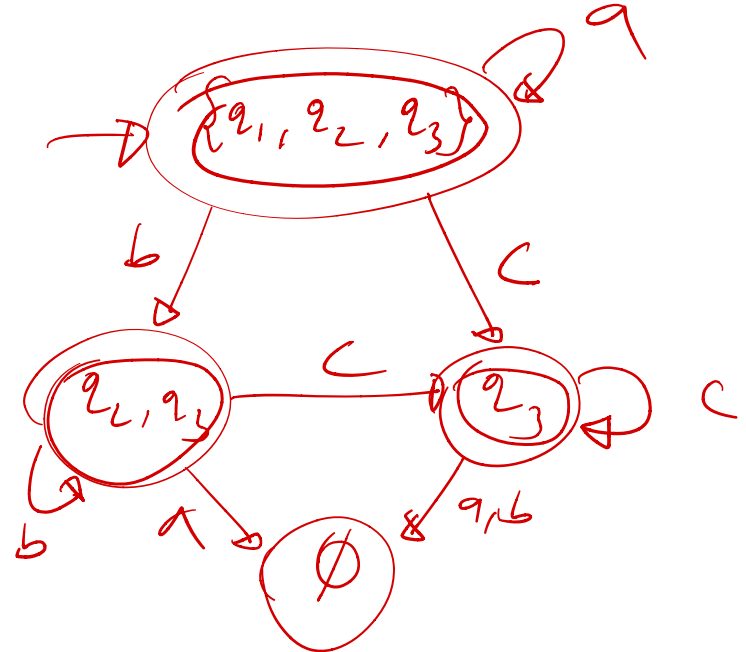


$$\varepsilon\text{-closure}(q_1) = \{q_1, q_2, q_3\}$$

$$\varepsilon\text{-closure}(q_2) = \{q_2, q_3\}$$

$$\varepsilon\text{-closure}(q_3) = \{q_3\}$$

	a	b	c
$\{q_1, q_2, q_3\}$	$\{q_1, q_2, q_3\}$	$\{q_2, q_3\}$	$\{q_3\}$
$\{q_2, q_3\}$	$\emptyset$	$\{q_2, q_3\}$	$\{q_3\}$
$\{q_3\}$	$\emptyset$	$\emptyset$	$\{q_3\}$



Recap

Regular sets of strings i.e., those that aren't too complicated are those that are

- the language of some DFA, or equivalently,
- the language of some NFA.

If a set of strings can be expressed as the result of **complement, union, intersection, concatenation, Kleene star** of regular languages, then it itself is regular.

NFA $\rightarrow$ ∞ $\checkmark$
 $\downarrow$
DFA $\rightarrow$ 2 $\checkmark$